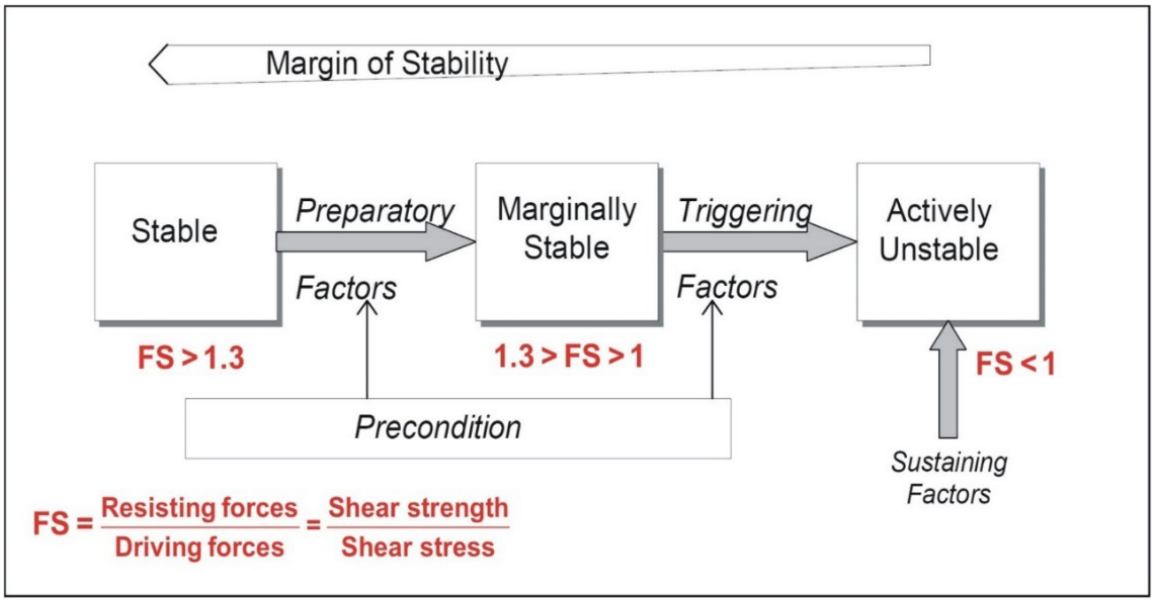


Figure 12: *Stability states and destabilising factors. Adapted from Glade and Crozier (2005). Landslides occur when the factor of safety (FS) is < 1.*

The most common causative processes that will have an impact of the changing characteristics of the factor of safety are listed in Table 13 and partly summarized in Figure 13. Combinations of (a) and (b) often act simultaneously as a direct result of external processes, as in the case of basal erosion or excavations, which can cause both an increase in shear stress through increased slope angle and/or height, or a decrease in shear strength, through a reduction in total and effective stress.

Table 13: **Triggering factors with examples of common causative/forcing processes. Adapted from Leroueil (2001) and Lu and Godt (2013).**

Triggering factor	Common causative/forcing processes
Decrease in shear strength	Infiltration due to rainfall, irrigation, leakage from utilities and the associated incrrease in pore-water presssures or changes in hydrolic conditions/properties
	Construction activities, e.g. pile diving
	Wheatering and cjnage is regolith or rock shear-strength properties
	Root decay (e.g. from forest removal)
Increase in shear stress	Weigth extra due to rianfall - weight of water
	Erosion or excavation at the toe
	Overloading at the top (e.g. by other mass movements or by placement of fill
	Mass redistribution (e.g. due to terracing)
	Transient loading from earthquake